

Muhammad Fikri Riyanto

BUSINESS ANALYST PORTOFOLIO

PT. 1

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ABOUT ME.



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Muhammad Fikri Riyanto

I'm a Mathematics student of Walisongo State Islamic University I'm an individual that is adaptable, responsible, hardworking, enjoy work in team and really eager to learn many of things related to technology especially in the data sciences with Python. I always looks for opportunities to develop my capacity, both in terms of soft skills and technical skills.

TABLE OF CONTENT

1

ABOUT ME

A summary about my profile and experiences

2

INTRODUCTION

business problem statement that i will solve in this case studies

3

PROBLEM BREAKDOWN

shattering business problem into small chunks so that further analysis can be easier

4

SOLUTION

description of solution that i provide to answer the business problem

INTRODUCTION

Scenario

Bluejay Natural Gas is a large energy company based out of Baltimore, MD. The company offers a wide range of energy products and has annual revenues of approximately \$50 billion. Because of the diverse nature of the company, its Manager for Project Development, Julie Kavoli, is under continual pressure to manage project proposals from the functional areas of the company. At any point in time, there might be dozens of projects at various stages requiring a wide variety of capital expenditures, promising widely varying future revenue streams, and containing varying degrees of risk. Julie has a difficult balancing act. The company's CEO, Cordelia Kareeni, is very concerned about keeping capital expenditures within a fixed budget and managing risk. The heads of the company's functional areas are less worried about budgets and risks; they are most concerned that their pet projects are approved. Julie also knows that many of the proposed projects, especially those requiring large capital expenditures, must be led by senior project managers with the appropriate experience and skills, and she is keenly aware that the company has only a limited supply of such managers.

Julie is currently preparing to meet with all parties involved to discuss project proposals for the next three years. She has proposals from the various functional areas for projects they would like to undertake. Each of these is accompanied by a schedule of capital expenditures over the next three years and a financial analysis of the expected revenue streams. These lead to a net present value (NPV) for each proposal, using the company's discount rate of 12%.

Several of the projects, if approved, must be undertaken in joint partnership with another company. For example, if project 3 is approved, Bluejay Natural Gas will take a 50% share in the project, and the other 50% will be shared by a partner. In this case, Bluejay Natural Gas will only incur 50% of the expenditures and receive only 50% of the revenues.

The table of financial estimates for potential projects (in millions of dollars) is contained in the attached Excel file.

KEYPOINT

- Annual revenues \$50 Billion
- Dozen of projects needs expendicture
- manage capital expendicture within a fixed budget
- satisfy the condition for each functional area
- assign senior project manager with suitable skills for each approved projects

DATA GIVEN:

- There is 12 projects divided by 3 functional area (FA), FA 1 has 5, FA 2 has 4, lastly FA 3 has 3
- Partnership percentage refers to company’s share of project with another company
- Capex column is the sum of capital expendicture for each projects in each year
- NPV is short for Net Profit Value and it shows estimation for NPV in each projects

Financial Estimates for Potential Projects (in \$millions)						
Project Index	Functional Area (FA)	Partnership Percentage	Capex Year 1	Capex Year 2	Capex Year 3	NPV
1	FA1	100%	250	100	100	60
2	FA1	33%	500	300	300	180
3	FA1	50%	100	200	400	80
4	FA1	100%	750	500	300	310
5	FA1	75%	200	400	800	220
6	FA2	50%	1000	300	300	180
7	FA2	100%	750	750	300	410
8	FA2	100%	800	700	600	280
9	FA2	67%	400	600	800	380
10	FA3	100%	100	200	400	100
11	FA3	50%	700	500	300	260
12	FA3	100%	1500	400	400	340

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PROJECT BREAKDOWN

Before the big meeting, Julie wants to be thoroughly prepared to answer all the questions she knows she will be asked by the CEO, the functional heads, and other interested parties. As a first step, she wants to develop an Excel spreadsheet model including the following information. Check off each item as you incorporate it into the model.

- ☒ The model should list the given financial information for all potential parties.
- ☒ The model should associate with each proposed project a cell that is 1 if the project is approved, 0 if not approved.
- ☐ For now, you can insert 0 or 1s randomly. Try to manually find a solution of projects to assign where all conditions are satisfied. Later in the final project we will optimize this.
- ☒ Each functional area wants as many of its projects to be approved as possible, but it certainly does not want to be shut out altogether. The model should indicate each of the following: the number of approved projects for each function area and whether each function area gets at least one approved project. The latter should be indicated by a 0/1 value: 0 if the function area is shut out, 1 if it gets at least one project approved.
- ☒ The model should find the sum of capital expenditures for all approved projects, both by year and by the total over all three years. It should address Cordelia's concerns by indicating whether the total for all three years exceeds \$10 billion and, for each year, whether that year's total expenditures exceed \$4 billion. As in the previous direction, these should be indicated by 0/1 values.
- ☒ The model should find the sum of capital expenditures for all approved projects, by year, for each functional area.
- ☒ Recognizing that the NPVs in the attached spreadsheet are probably optimistic estimates, the model should include another input, the possible percentage decrease in each original NPV. Then there should be a new NPV column that reflects this percentage decrease.
- ☒ The model should list the ROI (Return on investment) for each project, defined as the NPV as a percentage of the total capital expenditure for the project. Each ROI should be based on the adjusted NPV from the previous direction. The ROI for any project is the same, regardless of whether Bluejay undertakes it at the 100% level or in a partnership.
- ☒ The model should include a one-way data table in which the column input is the common percentage decrease in each NPV mentioned previously, varied from 0% to 30% in increments of 5%. The outputs in the data table should be the ROIs of undertaken projects: the average; the minimum; and the maximum. Of course, the results in this data table should update automatically if the 0/1 values in the second bullet are changed.
- ☒ The model should be accompanied by a line chart of total capital expenditures for all approved projects by year and a separate such line chart for each functional area.

after comprehending all of the requirement , i decided to divide the solution into four segments as described below

FOUR SEGMENTS

1. first segment will be financial information of each Functional Area projects. hence its consisted of :
 - Adjusted capital expendicture based on partnership percentage
 - Sum of capital expendicture grouped by year and total capital expendicture in each FA
 - New NPV adjusted from NPV discount rate (12%)
 - Add ROI (Return of Investment) column where the values will be automatically inputted if the project is approved
 - Approval rating column, the values will be 1 if approved 0 for declined
 - Lastly, total of capital expendicture combined
2. Second segment is the given information of Bluejay Gas company :
 - Annual revenue from the problem statement
 - Minimum capital expendicture for each FA in each year
 - Maximum total capital expendicture for three years
 - Sum of approved projects in each FA
3. Third segment will be Return of Investment one way table. this table will be composed by three types aggregation rows from the first segment ROI financial table.
 - Average
 - Maximum
 - MinimumWhile the column input will be common percentage decreaase in each NPV, varied from 0%-30% with the increments of 5%.
4. Last the fourth segment is a requirement checks table which consisted of the constraint
 - Capital expendicture in each year must be below \$4 Billion
 - Total Capital expendicture in three years time must be below \$10 Billion
 - Each FA must at least have one project
 - Requirement check for all condition is met

Expectation Input				
	Expendicture (Millions)	10.000		
	minimum exp (Millions)	4000		
	NPV Discount Rate Decrease	12%		
	FA1 Approved Project	5	=SUM(K18:K22)	
	FA2 Approved Project	4	=SUM(K25:K28)	
	FA3 Approved Project	3	=SUM(K31:K33)	

◀ SECOND

SUMMARY AND CONSTRAINT

Expectation and given input: a table consisted of given information and summary of numbers of approved projects in each Functional Area.

THIRD ▶▶

ONE WAY TABLE FOR PREDICTING ROI (RETURN OF INVESTMENT)

I'm using one way table to predict three section of ROI:

- a. Average : average of ROI in all approved projects
- b.Maximum: Maximum ROI in each approved projects
- c. Minimum: Minimum ROI in each approved projects

to do this the column input is the NPV discount rate in the second segment table

	ROI		
NPV Rate Change	Average	Maximum	Minimum
	21%	44%	12%
0%	23%	50%	13%
5%	22%	47%	13%
10%	21%	45%	12%
15%	20%	42%	11%
20%	19%	40%	11%
25%	18%	37%	10%
30%	16%	35%	9%

FOURTH: REQUIREMENT CHECKS

Requirement Checks					
Capital yr 1 <4000	0	=IF(SUM(D23;D29;D34)<=4000;1;0)			
Capital yr 2 <4000	1	=IF(SUM(E23;E29;E34)<=4000;1;0)			
Capital yr 3 <4000	1	=IF(SUM(F23;F29;F34)<=4000;1;0)			
Total Cap <10000	0	=IF(SUM(G23;G29;G34)<10000;1;0)			
FA at least has 1 project	1	=IF(AND(O5>=1;O6>=1;O7>=1);1;0)			
Are all Condition met?	0	=IF(SUM(O23:O27)=5;1;0)			

The last segment will serve as requirement checks. the table consist of 2 columns;

- 1.constraint information text
- 2.rating number: 1 if it passed, 0 if it not, also the color of the cells will changes depends on the rating number using conditional formatting.